MILITARY SPECIFICATION SHEET

ELECTRON TUBE, TRANSMITTING

TYPE 809

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Triode, amplifier, F1 = 60 MHz, F2 = 100 MHz

Outline	---	See figure 1
Cathode	---	Thoriated tungsten filament

Base connections:

Pin No.	---	1	2	3	4	Cap
Element	---	f	nc	g	f	a

ABSOLUTE-MAXIMUM RATINGS:

Parameter: / Unit:	Ef Vdc	Eb Vdc	Ec Vdc	Ib mAdc	Ic mAdc	Pp W	Pi W	Modulation ---	Alt ft
Maximum:									
Class B, AF:	6.6	750	---	100	---	25	75	---	10,000
Class B, Telep:	6.6	750	---	50	---	25	37.5	---	10,000
Class C, Telep:	6.6	600	-200	83	35	17.5	50	Anode	10,000
Class C, Teleg:	6.6	750	-200	100	35	25	75	---	10,000
Minimum:	6.0	---	---	---	---	---	---	---	---
TEST CONDITIONS:	6.3	1,000	---	---	---	---	---	---	---

GENERAL:

Qualification - Required

Ⓐ denotes changes

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Qualification inspection							
1236	~~Power oscillation~~ (1)	Power oscillation (2), except F = 60 MHz	---	---	Po	50	---	W
A 1216	Base material insulating quality							
	Quality conformance inspection, part 1							
1236	Power oscillation (2)	Eb = 100 mAdc; Ic = 35 mAdc; Rg = 2,000 ohms; F = 15 MHz (see note 2)	0.65	II	Po	55	---	W
1231	Emission by oscillation	Power oscillation (2) (see note 1)	0.65	II	Ef	---	5.7	Vdc
1256	Electrode current (anode)	Ec = -8 Vdc	0.65	II	Ib	15	25	mAdc
1266	Total grid current	Ec Ib = 25 mAdc (see note 1)	0.65	II	Ic	0	-5.0	μAdc
1301	Filament current		0.65	II	If	2.37	2.63	Adc
	Quality conformance inspection, part 2							
1031	Low-frequency vibration	Eb = 300 Vdc; Ec = 0; Rp = 2,000 ohms	---	---	Ep	---	500	mVac
1036	Bump	Angle = 20°	---	---	---	---	---	---
1236	Internal insulation		---	---	---	---	---	---
1316	Amplification factor	Eb = 800 Vdc; Ec/Ib = 20 mAdc	---	---	Mu	50	60	---
1331	Direct-interelectrode capacitance		---	---	Cgp Cgk Cpk	5.8 4.7 0.6	7.5 6.7 1.2	pF pF pF
A 1101	Secureness of base, cap, or insert		---	---	---	---	---	---
A 1105	Permanence of marking		---	---	---	---	---	---
	Quality conformance inspection, part 3							
---	Life-test provisions	Group C; Pp = 25 W	---	---	---	---	---	---
---	Life-test end point (500 hours)	Emission by oscillation	---	---	Ef	---	6.0	Vdc

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NOTES:

1. This test to be performed at the conclusion of the holding period.
2. This test may be made at 5 MHz, or above, provided correlative Q (figure of merit) tests are made at 60 MHz.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - EL
Navy -
Air Force - 11, 80
DSA - ES

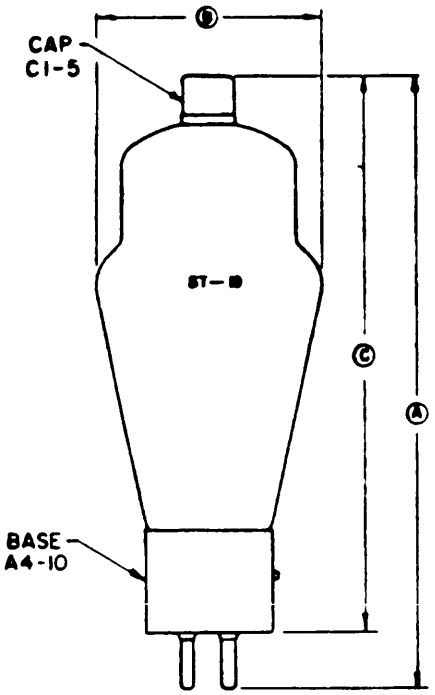
User activities:
Army - MC
Navy - AS, OS, CG, SH
Air Force - 19

Preparing activity: Navy - EC

Agent: DSA - ES

(Project 5960-2425-59)

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	6.25 (158.75)	6.56 (166.62)
B		2.44 (61.98)
C	5.65 (143.51)	

FIGURE 1. Outline drawing of electron tube type 809.

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801

C-26384

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